Module 6.2 – Case Study: Blackboard Learn

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Blackboard Inc., a major educational tech provider, started facing escalating technical debt in its flagship Learn product. Dating back to 1997, its product was supported with a J2EE monolith with legacy Perl language fragments still embedded into the code, and warning signs started to show as the company product needed to scale.

Repository data from 2005 onward started to reveal these signs years before action was taken, as code commits in ~2009 started to drop sharply and stay flat, even as the monolith’s lines of code continued increasing until 2012’s change. The gap between the rising code lines and falling commit changes was used as evidence that the system was becoming harder to modify, leading to technical debt and in worst case, architectural decay. This piece of evidence was what prompted then-chief architect David Ashman to pursue a rewrite.

Thus, in 2012, the team applied the strangler fig pattern via “Building Blocks”, which were modular components that were decoupled from the monolith, accessed only through fixed APIs, and let developers work independently and autonomously. Interestingly, this had existed since 2005, but was small, barely-used until it was pushed for adoption. With this push, developers started migrating their work, with commits and lines of code both increasing, well actually, exploding in 2012, while the monolith’s lines of code finally started shrinking within that same window period. This was pretty much a direct piece of evidence that the migration was in full progress.

It shows the frequency of commits and amount of lines of code can be used as an indicator of this potential technical debt and decay. It also interestingly puts focus on development teams, as this migration was in part successful by letting developers move autonomously, rather than mandate it, which drove the push of change to adapt. May it be said, all without halting new feature production, like a certain networking social media company did with their monolith.